1. File and Directory Management

- **ls** – List directory contents
- **cd** – Change directory
- **pwd** – Print working directory
- **cp** – Copy files and directories
- **mv** – Move or rename files and directories
- **rm** – Remove files or directories
- **mkdir** – Make directories
- **rmdir** – Remove empty directories
- **touch** – Change file timestamps or create empty files
- **find** – Search for files in a directory hierarchy
- **locate** – Find files by name
- **tree** – Display directories in a tree -like format
- **chmod** – Change file permissions
- **chown** – Change file owner and group
- **chgrp** – Change group ownership
- **stat** – Display file or file system status

2. File Viewing and Editing

- **cat** – Concatenate and display file content
- **tac** – Concatenate and display file content in reverse
- **more** – View file content interactively (page by page)
- **less** – View file content interactively (scrollable)
- **head** – Output the first part of a file
- **tail** – Output the last part of a file
- **nano** – Text editor (terminal -based)
- **vim / vi** – Advanced text editors
- **emacs** – Text editor
- **grep** – Search text using patterns
- **sed** – Stream editor for filtering and transforming text
- **awk** – Pattern scanning and processing langu age
- **cut** – Remove sections from each line of files
- **sort** – Sort lines of text files

- **uniq** – Report or omit repeated lines

3. Process Management

■ **ps** – Report a snapshot of current processes
■ **top** – Display Linux tasks
■ **htop** – Interactive process viewer (advanced top)
■ **kill** – Send a signal to a process, typically to terminate
■ **killall** – Terminate processes by name
■ **bg** – Resume a suspended job in the background
■ **fg** – Bring a job to the foreground
■ **jobs** – List active jobs
■ **nice** – Run a program with modified scheduling priority
■ **renice** – Alter priority of running processes
■ **uptime** – Show how long the system has been running
■ **time** – Measure program running time

4. Disk Management

■ **df** – Report file system disk space usage
■ **du** – Estimate file space usage
■ **fdisk** – Partition table manipulator for Linux
■ **lsblk** – List information about block devices
■ **mount** – Mount a file system
■ **umount** – Unmount a file system
■ **parted** – A partition manipulation program
■ **mkfs** – Create a file system
■ **fsck** – File system consistency check and repair
■ **blkid** – Locate/print block device attributes

5. Networking

■ **ifconfig** – Configure network interfaces
■ **ip** – Show/manipulate routing, devices, and tunnels
■ **ping** – Send ICMP Echo requests to network hosts

■ **netstat** – Network statistics
■ **ss** – Socket statistics (faster than netstat)
■ **traceroute** – Trace the route packets take to a network host
■ **nslookup** – Query Internet name servers interactively
■ **dig** – DNS lookup utility
■ **wget** – Non-interactive network downloader
■ **curl** – Transfer data with URLs
■ **scp** – Secure copy files between hosts

■ **ssh** – Secure shell for remote login
■ **ftp** – File Transfer Protocol client

6. User and Group Management

■ **useradd** – Add a user to the system
■ **usermod** – Modify a user account
■ **userdel** – Delete a user account
■ **groupadd** – Add a group to the system
■ **groupdel** – Delete a group
■ **passwd** – Change user password
■ **chage** – Change user password expiry information
■ **whoami** – Print the current logged -in user
■ **who** – Show who is logged in
■ **w** – Show who is logged in and what they're doing
■ **id** – Display user and group information
■ **groups** – Show user's groups

7. System Information and Monitoring

■ **uname** – Print system information
■ **hostname** – Show or set the system's hostname
■ **uptime** – How long the sys tem has been running
■ **dmesg** – Boot and system messages
■ **free** – Display memory usage
■ **top** – Display Linux tasks
■ **vmstat** – Report virtual memory statistics

■ **lscpu** – Display information about the CPU architecture
■ **lsusb** – List USB devices
■ **lspci** – List PCI devices
■ **lshw** – List hardware configuration

8. Archiving and Compression

■ **tar** – Archive files
o **tar -czf archive.tar.gz /path/to/directory** – Compress files using gzip
o **tar -xzf archive.tar.gz** – Extract gzipped tarball
o **tar -cf archive.tar /path/to/directory** – Create a tarball
o **tar -xf archive.tar** – Extract tarball
■ **zip** – Package and compress files into a ZIP archive

- **unzip** – Extract files from a ZIP archive
- **gzip** – Compress files using the gzip algorithm
- **gunzip** – Decompress files compressed with gzip
- **bzip2** – Compress files using the bzip2 algorithm
- **bunzip2** – Decompress files compressed with bzip2
- **xz** – Compress files using the xz algorithm
- **unxz** – Decompress files compressed with xz

9. Package Management (Depends on Distribution)

Debian -based (e.g., Ubuntu)

- **apt-get** – APT package handling utility
  - **apt-get install** – Install a package
  - **apt-get update** – Update package list
  - **apt-get upgrade** – Upgrade installed packages
  - **apt-get remove** – Remove a package
- **apt-cache** – Query APT cache
  - **apt-cache search** – Search for a package
  - **apt-cache show** – Show package details

Red Hat -based (e.g., CentOS, Fedora)

- **yum** – Package manager for RPM -based systems
  - **yum install** – Install a package
  - **yum update** – Update installed packages
  - **yum remove < package>** – Remove a package
- **dnf** – Next -generation package manager (Fedora, CentOS 8+)
  - **dnf install** – Install a package
  - **dnf update** – Update installed packages
  - **dnf remove** – Remove a package

General Commands

- **rpm** – RPM package manager
  - **rpm -i** – Install an RPM package
  - **rpm -e** – Remove an RPM package
- **dpkg** – Debian package manager
  - **dpkg -i** – Install a Debian package
  - **dpkg -r** – Remove a Debian package

10. System Services and Daemon Management

- **systemctl** – Control the systemd system and service manager

o **systemctl start** – Start a service
o **systemctl stop** – Stop a service
o **systemctl restart** – Restart a service
o **systemctl enable** – Enable a service to start on boot
o **systemctl disable** – Disable a service from starting
on boot
o **systemctl status** – Check service status
■ **service** – Older service management command (used in non -
systemd systems)
o **service start** – Start a service
o **service s top** – Stop a service
o **service restart** – Restart a service
o **service status** – Check service status

11. Scheduling Tasks
■ **cron** – Daemon for running scheduled commands
o **crontab -e** – Edit cron jobs for the current user
o **crontab -l** – List the current user's cron jobs
o **crontab -r** – Remove the current user's cron jobs
■ **at** – Run commands at a specified time
o **at 09:00** – Schedule a command to run at 09:00 AM
■ **batch** – Run commands when the system load is low
■ **sleep** – Delay for a specified time
o **sleep 5s** – Sleep for 5 seconds

12. File Permissions and Security
■ **chmod** – Change file permissions
■ **chown** – Change file owner and group
■ **chgrp** – Change the group ownership of a file
■ **umask** – Set default permissions for new files
■ **setfacl** – Set file access control lists (ACL)
■ **getfacl** – Get file access control lists (ACL)
■ **sudo** – Execute a command as another user (usually root)
■ **visudo** – Edit the sudoers file safely
■ **passwd** – Change a user's password
■ **sudoers** – Manage sudo access for user s
■ **gpasswd** – Administer group password
■ **ss** – Display socket statistics (for secure network connections)

13. System Backup and Restore
■ **rsync** – Remote file and directory synchronization
o **rsync -avz source/ destination/** – Synchronize files
o **rsync -avz -e ssh source/ user@remote:/destination/** – Sync over SSH
■ **cpio** – Copy files to and from archives
■ **dd** – Low-level copying and backup of entire filesystems

o **dd if=/dev/sda of=/path/to/backup.img** – Backup a disk/partition
o **dd if=/path/to/backup.img of=/dev/sda** – Restore a disk/partition
14. System Diagnostics and Troubleshooting
■ **dmesg** – Print the kernel ring buffer messages (system boot and hardware -related messages)
■ **journalctl** – Query and view logs from systemd's journal
■ **strace** – Trace system calls and signals
o **strace** – Trace a command's system calls
■ **lsof** – List open files (useful for debugging)
o **lsof** – Show processes using a specific file
■ **vmstat** – Report virtual memory statistics
■ **iostat** – Report CPU and I/O statistics
■ **mpstat** – Report CPU usage statistics
■ **pidstat** – Report statistics by process
■ **free** – Display memory usage
■ **uptime** – How long the system has been running
■ **watch** – Execute a program periodically, showing output
o **watch -n 1 free** – Watch memory usage every second
■ **lshw** – List hardware configuration
■ **htop** – Interactive process viewer (better than top)
■ **netstat** – Network statistics (deprecated in favor of ss)
■ **ss** – Show socket statistics (more efficient than netstat)

15. Networking & Remote Management

■ **ifconfig** – Configure netw ork interfaces (older command, replaced by ip)
■ **ip** – A more modern alternative for managing network interfaces

and routing
o **ip addr** – Show IP addresses
o **ip link** – Show or manipulate network interfaces

o **ip route** – Show or manipulate routing tables
■ **ss** – Display socket statistics (useful for diagnosing network issues)
■ **nmap** – Network exploration tool (can be used for security auditing)
■ **telnet** – User interface to the TELNET protocol (less common nowadays)
■ **nc (Netcat)** – Network utility for reading and writing from network connections
o **nc -l -p 1234** – Listen on port 1234
o **nc** – Connect to a host and port
■ **iptables** – Administration tool for IPv4 packet filtering and NAT (Network Address Translation)
■ **firewalld** – Frontend for managing firewall rules (used in some distros like Fedora and CentOS)
■ **ufw** – Uncomplicated firewall (front -end for iptables)
o **ufw enable** – Enable firewall
o **ufw allow** – Allow traffic on a specific port
■ **tcpdump** – Command -line packet anal yzer
■ **curl** – Transfer data from or to a server using various protocols (HTTP, FTP, etc.)
■ **wget** – Download files from the web via HTTP, HTTPS, FTP
■ **scp** – Secure copy over SSH (used to copy files between systems)
o **scp file.txt user@remote:/path/to/destination/** – Copy file to remote server
■ **rsync** – Remote file and directory synchronization (often used for backups)
o **rsync -avz /local/path/ remote:/remote/path/** – Sync directories

16. Text Processing Utilities

■ **grep** – Search for patterns within files
o **grep 'pattern' file.txt** – Search for a pattern in a file

o **grep -r 'pattern' /dir/** – Recursively search for a pattern

■ **sed** – Stream editor for filtering and transforming text

o **sed 's/old/new/g' file.txt** – Replace old with new globally

■ **awk** – A powerful text processing langua ge

o **awk '{print $1}' file.txt** – Print the first column of each line in a file

■ **cut** – Remove sections from each line of a file

o **cut -d ':' -f 1 /etc/passwd** – Print the first field of each line, delimited by ":"

■ **sort** – Sort lines of text files

o **sort file.txt** – Sort file content in ascending order

■ **uniq** – Report or omit repeated lines in a file

o **sort file.txt | uniq** – Sort and remove duplicate lines

■ **tee** – Read from standard input and write to standard output and files

o **echo "text" | tee file.txt** – Write to file and show output on screen

■ **tr** – Translate or delete characters

o **echo "hello" | tr 'a -z' 'A-Z'** – Convert lowercase to uppercase

■ **paste** – Merge lines of files

o **paste file1.txt file2.txt** – Combine lines of file1 and file2 side by side

■ **wc** – Word, line, character, and byte count

o **wc -l file.txt** – Count lines in a file

o **wc -w file.txt** – Count words in a file

17. System Shutdown and Reboot

■ **shutdown** – Shut down the system

o **shutdown -h now** – Immediately shut down

o **shutdown -r now** – Reboot the system

o **shutdown -h +10** – Shut down after 10 minutes

■ **reboot** – Reboot the system

■ **halt** – Halt the system immediately (equivalent to turning off power)

■ **poweroff** – Power off the system

■ **init** – Change the runlevel (old -style system manager)

o init 0 – Shutdown

o init 6 – Reboot

18. File System Mounting and Management

■ **mount** – Mount a file system

o mount /dev/sda1 /mnt – Mount partition to a directory

■ **umount** – Unmount a file system

o umount /mnt – Unmount the file system mounted at /mnt

■ **fstab** – File system table (configuration file for mounting file systems)

o /etc/fstab – View and configure persistent mount points

■ **blkid** – Display block device attributes

■ **fsck** – Check and repair a file system

o fsck /dev/sda1 – Check and repair /dev/sda1

19. Filesystem Permissions and Security

■ **chmod** – Change file permissions

o chmod 755 file.txt – Give read, write, and execute permissions to owner, and read -execute permissions to others

■ **chown** – Change file owner and group

o chown user:group file.txt – Change own er and group of a file

■ **chgrp** – Change group ownership of a file

o chgrp group file.txt – Change the group of a file

■ **umask** – Set default permissions for new files

o umask 022 – Set default permissions for newly created files to 755

■ **setfacl** – Set access contro l lists (ACL) for file permissions

■ **getfacl** – Get access control lists (ACL) for file permissions

20. Containerization and Orchestration

Docker

■ **docker** – Docker command -line interface (CLI) for managing containers

o docker run – Run a container from an image

o docker ps – List running containers

o docker ps -a – List all containers, including stopped ones

o docker build -t . – Build an image from a Dockerfile

o **docker exec -it bash** – Start an
interactive bash shell inside a running container

o **docker stop** – Stop a container

o **docker rm** – Remove a container

o **docker logs** – View logs of a container

o **docker images** – List available images

o **docker rmi** – Remove an image

o **docker network ls** – List Docker networks

o **docker-compose** – Manage multi -container Docker
applications

■ **docker-compose up** – Start up a multi -container
environment

■ **docker-compose down** – Stop and remove containers
created by docker -compose

■ **docker-compose logs** – View logs from co ntainers
managed by docker -compose

Kubernetes (k8s)

■ **kubectl** – Command -line tool for interacting with Kubernetes
clusters

o **kubectl get pods** – List pods in the current namespace

o **kubectl get nodes** – List nodes in the cluster

o **kubectl get services** – List services in the cluster

o **kubectl apply -f .yaml** – Apply configuration
from a file (e.g., a deployment or pod configuration)

o **kubectl create -f .yaml** – Create a resource
from a file

o **kubectl delete -f .yaml** – Delete a resource
defined in a fi le

o **kubectl exec -it -- bash** – Execute a
command inside a pod (e.g., open a shell)

o **kubectl logs** – View the logs of a pod

o **kubectl describe pod** – Get detailed
information about a pod

o kubectl scale deployment --
replicas= – Scale a deployment to the desired

number of replicas
o kubectl rollout restart deployment
– Restart a deployment
o kubectl port -forward pod
: – Forward a port from a
pod to localhost
Helm
■ **helm** – Kubernetes package manager for deploying applications
o helm install –
Install a Helm chart
o helm upgrade –
Upgrade a Helm release
o helm list – List all Helm releases
o helm delete – Delete a Helm release
o helm search – Search for a Helm chart

21. Automation and Configuration Management

Ansible
■ **ansible** – Automation tool for configuration management

o ansible all -m ping – Ping all hosts defined in the
inventory
o ansible-playbook playbook.yml – Run an Ansible
playbook
o ansible -m command -a 'command' – Run a
single command on a target host
o ansible-playbook --check playbook.yml – Dry-run
a playbook to see what would change
o ansible-playbook --limit playb ook.yml –
Run a playbook on a specific host or group
o ansible-playbook --extra-vars "key=value" –
Pass extra variables to a playbook
Terraform
■ **terraform** – Infrastructure as code tool for provisioning and
managing cloud resources
o terraform init – Initialize a working directory for
Terraform configuration

o **terraform plan** – Show an execution plan (preview of what changes will be made)

o **terraform apply** – Apply the changes described in a Terraform configuration

o **terraform destroy** – Destroy infrastructure created by Terraform

o **terraform validate** – Validate the configuration files

o **terraform show** – Show the current state of the infrastructure

Puppet

■ **puppet** – Configuration management tool

o **puppet apply** – Apply a Puppet manifest locally

o **puppet agent --test** – Test the Puppet agent (can be used to run a one -off run)

o **puppet resource** – Show the current state of resources (files, services, etc.)

22. CI/CD Tools and Commands

Jenkins

■ **jenkins** – Continuous integration tool

o **java -jar jenkins.war** – Start Jenkins from a WAR file

o Access Jenkins through http://localhost:8080 by default

GitLab CI

■ **.gitlab-ci.yml** – Configuration file for GitLab CI/CD pipelines (typically resides in your repository)

o **gitlab-runner register** – Register a new runner with GitLab

o **gitlab-runner run** – Run the GitLab Runner to process jobs

GitHub Actions

■ GitHub Actions uses YAML configuration files (typically located in .github/workflows/)

o **actions/checkout@v2** – Checkout the repository code in your CI pipeline

o **actions/setup -node@v2** – Setup Node.js for use in a pipeline

o docker/setup -buildx-action@v1 – Set up Docker Buildx for building multi -platform images

23. Cloud Services

AWS CLI (Amazon Web Services)

■ **aws** – Command -line tool for managing AWS services

o aws configure – Configure AWS CLI with your credentials

o aws s3 cp file.txt s3://bucket -name/ – Copy a file to an S3 bucket

o aws ec2 describe -instances – Describe EC2 instances

o aws ec2 start -instances --instance -ids – Start an EC2 instance

o aws ec2 stop -instances --instance -ids – Stop an EC2 instance

o aws s3 sync – Sync directories with an S3 bucket

Azure CLI (Microsoft Azure)

■ **az** – Command -line tool for managing Azure services

o az login – Log in to your Azure account

o az vm list – List all virtual machines

o az vm start --name --resource -group – Start an Azure VM

o az storage blob upload – Upload files to an Azure blob storage

o az group create – Create a new resource group in Azure

Google Cloud SDK (gcloud)

■ **gcloud** – Command -line tool for Google Cloud Platform

o gcloud auth login – Log in to Google Cloud

o gcloud compute instances list – List compute instances

o gcloud compute instances stop – Stop a Google Cloud VM instance

o gcloud app browse – Open the current Google App Engine application in a browse r

24. Logging and Monitoring

Prometheus

■ **prometheus** – Open -source system monitoring and alerting toolkit

o **prometheus** – Start Prometheus server (usually runs as a service in the background)

o **prometheus --config.file=** – Start Prometheus with a specific config file

Grafana

■ **grafana-cli** – Command -line interface for managing Grafana plugins

o **grafana-cli plugins install** – Install a plugin in Grafana

ELK Stack (Elasticsearch, Logstash, Kibana)

■ **elasticsearch** – Search engine for logging an d data analytics

o curl -XGET **'localhost:9200/_cluster/health?pretty'** – Get cluster health status

■ **logstash** – Server -side data processing pipeline

o **logstash -f** – Run Logstash with the specified configuration file

■ **kibana** – Web interface for visu alizing Elasticsearch data

o Kibana is generally accessed through a web browser (http://localhost:5601)